3D Brick & Block Studio

User guide · IngeTrazo plugin · `brick_block_studio.py` · Ctrl+Shift+T

Covers a selected wall face in 3D bricks or blocks, cutting them neatly around window and door openings, with bonds, header courses and soldier courses.

Installing and updating

1. Open File Explorer, paste `C:\Users\eddy_\AppData\Roaming\ingetrazo\plugins` into the address bar and press Enter.
2. Copy `brick_block_studio.py` into that folder. When updating, choose **Replace the file**.
3. Restart IngeTrazo. Settings are remembered only while IngeTrazo is open, so a restart also clears old settings.

Keep only one copy of the plugin in the folder. Two copies under different names may both load.

Shortcut: Ctrl+Shift+T

Using it

1. Select the wall face.
2. Press **Ctrl+Shift+T**, choose your settings and click OK.

If you select several faces, the one with the most openings is used as the wall. The other selected faces mark which openings get a soldier course: select the face filling an opening along with the wall.

Sizes and presets

Every size box shows its real value in the wall's units (mm or m). The defaults are a 210 × 65 × 20 mm brick slip with 10 mm joints.

Built-in preset	Length × height × depth (mm)
Brick slip	210 × 65 × 20
UK standard brick	215 × 65 × 102.5
German NF brick	240 × 71 × 115
US modular brick	194 × 57 × 92
Concrete block	440 × 215 × 100
Concrete block	390 × 190 × 190

Save as stores your own sizes under a name; **Delete** removes one of your presets. Your presets are kept in the file `.brick_block_studio_presets.json` in your user folder.

Bonds and row offsets

- **Bond pattern:** running (half offset), stack, third, English or Flemish.
- **Row offsets:** follow the bond, use a fixed offset on every second course, repeat a custom sequence of fractions such as `0, 0.5, 0.25, 0.75`, or randomise each course. Random offsets use a seed, so the same seed gives the same wall, and they avoid joints lining up with the course below.
- **Header courses:** add a course of headers every Nth course, with its own header width.

Openings and soldier courses

Tick **Soldier course around openings** to stand bricks on end around each opening. The list shows every opening in the wall with its size and position; tick the ones that get a course, or use **All** and **None**. **Soldier length** follows the unit length until you change it. **Corner angle** sets how sharp a bend must be to count as a corner.

Other settings

Setting	Effect
Mortar joint	Gap between bricks; 0 means no joint
Min cut fragment	Cut pieces smaller than this are left out
Max subdivisions	Detail used when cutting around curved or concave openings
Include back faces	Closes the back of every brick, so each one is a closed solid
Swap width / height axes	Lays the courses the other way across the face
Put units in a new group	Keeps the bricks separate from the wall

3D printing

The bricks need a backing, otherwise the slicer fills the gaps across the joints with V shapes. Either tick **Include back faces**, or duplicate the wall face and group it with the bricks; both slice cleanly.

At small scales, joints shrink below what a nozzle can print: a 10 mm joint is 0.2 mm at 1:50, half a standard 0.4 mm nozzle. Widen joints to 15–30 mm and deepen bricks before scaling, print larger, or use a 0.2–0.25 mm nozzle. Check the slicer's sliced layer preview to see what will actually print.

Troubleshooting

Problem	Cause	Fix
V shapes or slivers in mortar joints in the slicer	Bricks open at the back	Tick **Include back faces**, or group a backing face with the bricks
Mortar joints vanish when printed	Joints thinner than the nozzle	Widen joints before scaling, print larger, or use a finer nozzle
"No Face Selected" warning	Nothing, or a group, was selected	Select the flat wall face itself